AI-POWERED CYBERSECURITY

Assessment 01 • Task C: Triage a sign-in alert

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Task: C

Report date: 26 October 2026
Incident times: 18 October 2026, IST (UTC+05:30)

Scope and evidence

This assessment uses only the supplied synthetic sign-in log, classroom egress note and student statement [1]. No real account, system or IP address was accessed or tested. All source IP addresses are documentation examples. The usual classroom egress is 192.0.2.10; a different IP is a contextual clue, not proof of an attacker or a person’s identity.

Source log and statement

Working reconstruction of the supplied log

The pasted table has flattened columns. The following reconstruction reads each event and result in chronological order; the original table should be checked before treating this transcription as definitive. Every row concerns student07.

Time (IST)	Source IP	Event	Result
09:00	192.0.2.10	Password	Success
09:01	192.0.2.10	MFA	Approved
10:14	198.51.100.23	Password	Failed
10:15	198.51.100.23	Password	Failed
10:16	198.51.100.23	Password	Success
10:17	198.51.100.23	MFA	Denied
10:18	198.51.100.23	MFA	Approved
10:20	198.51.100.23	Session	Created

User statement: “I did not sign in after 09:05. At around 10:18, I approved a prompt because I thought it would stop the repeated requests.” [1]

C1. Four-point timeline

8 marks • Observations and interpretation are separated below. [1]

1. 09:00–09:01 — Classroom baseline

Observed: Password success at 09:00 and MFA approval at 09:01 are recorded for student07 from 192.0.2.10, the usual classroom egress. The student reports no sign-in after 09:05.

Interpretation: The early sequence is consistent with normal classroom use and does not conflict with the student’s statement. Familiar egress alone does not authenticate the person, and the excerpt does not record a session-creation event for this early sequence.

2. 10:14–10:16 — Failed attempts followed by success

Observed: Two password failures at 10:14 and 10:15 are followed by password success at 10:16, all from 198.51.100.23.

Interpretation: A successful password step after the student’s stated cut-off raises concern that someone else possessed or obtained the password. Two failures do not establish brute force, and the log does not show how the correct password became available. A legitimate client retry remains possible.

C1. Timeline continued

3. 10:17–10:18 — Denial followed by approval

Observed: MFA is denied at 10:17 and approved at 10:18 from the later source IP. The student acknowledges approving a prompt around 10:18 to stop repeated requests.

Interpretation: The statement corroborates the approval timing and supports suspected MFA fatigue or prompt bombing. The excerpt shows two MFA outcomes, not the total number of prompts received. Approval does not establish that the student intended to initiate or authorise the underlying sign-in.

4. 10:20 — Session created

Observed: A session is created for student07 from 198.51.100.23, two minutes after approval.

Interpretation: The order is consistent with the successful password and approved MFA enabling access. Correlation identifiers are needed to prove that these records belong to one authentication flow. No post-session actions, data access or exfiltration are shown.

C2. Explanations and checks

7 marks • Working assessment: suspected account compromise; moderate confidence in the mechanism.

Likely explanation: password compromise and MFA fatigue

An unauthorised party may have obtained student07’s password, completed the password step at 10:16, and triggered MFA prompts. After a denial, the student approved a prompt to stop the requests, potentially enabling the 10:20 session. The user statement, successful password step and subsequent session are the strongest combined evidence. This fits the MFA fatigue pattern addressed by CISA [2]. Password theft, the identity of the initiator and malicious use of the session remain unproven.

Alternative explanation: legitimate client reauthentication

A previously opened application on the student’s own device may have retried authentication while using another network or VPN. The student could truthfully report no deliberate sign-in after 09:05 while a background client generated prompts. Earlier attempts could reflect stale or mistyped credentials. This explanation requires device and application evidence; the changed IP by itself neither proves nor rules it out.

C2. Additional checks

Additional check 1 — Correlate the complete authentication flow

Obtain the original identity-provider records around 09:00–10:30, including request/correlation IDs, session IDs, target application, authentication details, MFA challenge identifiers, device identifiers and user-agent data. Confirm the reconstructed event/result columns and link the 10:16 password success, 10:18 approval and 10:20 session.

Discriminating result: A linked flow on an unfamiliar device or application would strengthen the compromise hypothesis. A flow matching a known managed device and an expected application's reauthentication would favour the alternative. Device and user-agent fields are supporting evidence, not conclusive identity proof.

Additional check 2 — Verify device context and session activity

Interview the student through a trusted channel and review authorised device/application logs, VPN or network-change records and MFA notification history. Examine what the 10:20 session accessed, including account changes, new MFA registrations, downloads or forwarding rules where relevant.

Discriminating result: Confirmed background retries from the student's device with expected activity would support the alternative. Unrecognised activity or security-setting changes would support compromise. Lack of logged activity does not prove that a session was harmless.

Neither IP address identifies a person. Both are synthetic documentation addresses; live geolocation or reputation checks would not establish attribution in this exercise. [1]

C3. Prioritised response actions

6 marks • Recommendations for the authorised IT team; no response actions were performed.

Priority 1 — Contain access and preserve evidence immediately

Open an incident record and promptly revoke active sessions and refresh tokens for student07 using the platform’s supported controls. Temporarily restrict sign-in if needed while ownership is verified. Confirm that existing sessions can no longer access applications; a password reset alone may not terminate them.

In parallel, export the original sign-in, MFA, session and audit records, retain the alert and student statement, and record timezone, collection time, collector and response actions. Preserve read-only originals with hashes and access controls. Do not delete the account or logs. Take a rapid evidence snapshot where feasible, but do not delay urgent containment for a lengthy collection. Reversible containment alongside evidence preservation is consistent with incident-response guidance [3].

Priority 2 — Recover and strengthen the account

Verify the student through a trusted channel, then reset the password to a unique value using a trusted device. Review registered MFA methods, recovery details and application grants; remove unauthorised additions after recording them. Re-enrol MFA if its registration or recovery mechanisms may be compromised. Restore access only after account control is verified.

Explain that unexpected prompts should be denied and reported. Prefer phishing-resistant MFA; if deployment cannot happen immediately, number matching is an interim improvement for push MFA [2]. Confirm that any new authentication method is securely enrolled.

C3. Investigation and recovery

Priority 3 — Establish impact and monitor recovery

Perform the two C2 checks, trace the 10:20 session and determine whether account settings or data were accessed or changed. Expand the review to related accounts only if evidence warrants it. Record confirmed findings separately from hypotheses and escalate under the organisation’s incident process if unauthorised access or data exposure is established.

Monitor renewed password attempts, unusual session creation and MFA requests after recovery. Close the incident only when ownership is restored, suspicious access is contained, the available evidence has been assessed and the student has clear reporting instructions.

Why this order?

Session creation makes potential continuing access the immediate concern. Containment limits exposure while evidence is retained. Account recovery reduces recurrence; impact investigation then establishes what happened and whether further response is required. These are overlapping workstreams, not reasons to postpone urgent protection.

C4. Incident summary

4 marks • 88 words

On 18 October 2026, student07 had a classroom sign-in, followed by two password failures, password success, MFA denial, MFA approval and session creation from another source IP. The student denies initiating a later sign-in but acknowledges approving a prompt to stop repeated requests. This supports suspected MFA fatigue and possible account compromise. The IP does not identify an attacker; background reauthentication remains an alternative. Treat as high priority: preserve authentication evidence while revoking sessions, then correlate request and device records, secure the account and investigate subsequent session activity.

References

[1] Assessment 01, Task C: “Triage a sign-in alert.” User-provided synthetic log, classroom egress note, student statement and assessment requirements. Incident date: 18 October 2026. Primary evidence for C1–C4.

[2] CISA. “CISA Releases Guidance on Phishing-Resistant and Numbers Matching Multifactor Authentication.” Guidance announcement, 31 October 2022. Supports the MFA fatigue mitigation recommendations; it is not evidence that this synthetic incident was an attack. Accessed 26 October 2026. https://content.govdelivery.com/accounts/USDHSCISA/bulletins/3355d59

[3] Amazon Web Services. “Contain.” AWS Security Incident Response User Guide. Supports reversible containment with evidence preservation; the lab does not specify an AWS environment. Accessed 26 October 2026. https://docs.aws.amazon.com/security-ir/latest/userguide/contain.html